

CLIMATE CHANGE

How Climate Change Is Hurting Transit Ridership

Transit isn't only a key solution to confronting climate change; it's also one of its victims.



By **Kea Wilson**



12:03 AM EDT on March 12, 2025



Photo: Freerange Stock

Climate-saving transit trips are dropping even as climate change makes weather worse, a new study finds — and to stop the vicious cycle, America needs to

better climate-proof its transit networks and encourage more people to choose shared modes now.

A recent study of 43 major U.S. transit networks revealed that ridership dropped during long stretches of rain, heat, or other prolonged extreme weather events that scientists link to climate change, at least in the 17 years between when the National Transit Database began collecting stats and 2019, the last year before the pandemic decimated ridership. That held true even when researchers controlled for what constitutes “extreme” weather in climates as diverse as Philadelphia and Phoenix, as well as how baseline transit ridership tends to differ depending on the race, education, and income mix of each community.

Of course, it’s not exactly news that no one likes to wait for the bus in a bad storm — and since people with no other options will ride regardless, the decline wasn’t very large. Still, the University of Oregon team behind the analysis warn that climate change-related ridership losses could get far worse, and that people are suffering and even dying at stops right now.

“Even now, you can end up waiting outside for the bus for quite a while, and that’s very uncomfortable in 90-degree heat,” said Nicole Ngo, the leader author on the paper. “And for certain populations — especially older adults — that can be dangerous, [even as] a lot of older adults use public transit as their primary mode of transportation, either due to mobility issues or due to income.”

Ngo explains that climate change doesn’t only tempt riders to skip transit trips on intemperate days — or even to take private vehicles instead. During bad snow and rain storms, buses can get snarled in the same traffic as cars, and rail tracks can warp in intense heat, forcing agencies to run trains more slowly or cancel trips outright, leaving even those who *would* pay the fare with no ride to take.

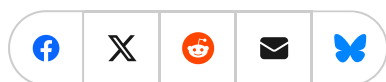
Worse, as extreme weather events drag on, ridership losses compound — especially during the harshest heat waves, which had a particularly marked effect in the latter half of the study period. And considering that transit is often low income residents’ only route to neighborhood cooling centers — and they can’t always afford to install or run air conditioning at home — that means the health impacts of heat are being compounded, too.

“It’s not just a matter of very hot days affecting travel behavior,” she added. “It’s much more pronounced when you have these *consecutive* days of heat or extreme weather.”

Ngo says apart from recognizing transit’s integral role in cutting emissions, transportation leaders need to confront the role of climate change itself in eroding ridership. That means not just improving and expanding service to make buses and trains a reasonable alternative to driving, but putting climate-controlling measures at stops, like shade, heat, A/C, cooling greenery, or anything else the climate requires — and doing so at a vast scale, considering that only about one-fifth of U.S. bus stops even have basic shelters now.

Ngo acknowledged that *neither* action will be easy, especially in a political climate where federal leaders are denying even the existence climate change. But she says it’s critical for them to accept the reality of extreme weather’s impact on transit, no matter what they believe is causing it.

“We need to realize that extreme weather is becoming much more common, and it’s not going away,” Ngo added. “We need to ensure that cities and transit agencies are aware that this is also hurting their bottom line. The effect is small now, but with heat waves and days with heavy precipitation increasing, the effects could grow over time.”



Kea Wilson

Kea Wilson is Senior Editor for Streetsblog USA. She has more than a dozen years experience as a writer telling emotional, urgent and actionable stories that motivate average Americans to get involved in making their cities better places. She is also a novelist, cyclist, and affordable housing advocate. She lives in St. Louis, MO. For tips, submissions, and general questions, reach out to her at kea@streetsblog.org, or on Bluesky [@keawilson.bsky.social](https://bsky.app/profile/keawilson.bsky.social).



Stay in touch

Sign up for our free newsletter

SIGN UP

Streetsblog has migrated to a new comment system. New commenters can register directly in the comments section of any article. Returning commenters: your previous comments and display name have been preserved, but you'll need to reclaim your account by clicking "Forgot your password?" on the sign-in form, entering your email, and following the verification link to set a new password — this is required because passwords could not be carried over during the migration. For questions, contact tips@streetsblog.org.

More from Streetsblog USA

AI

Where the Battles Against Harmful Highways and Hyper-Scale Data Centers Collide



Kea Wilson September 22, 2026

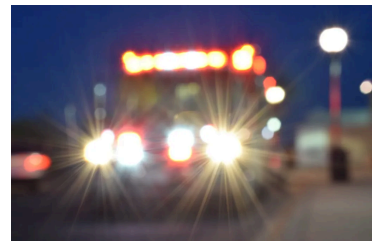


CAR CRASHES

I Survived the Santa Monica Farmer's Market Crash



Melissa H. Cronin September 22, 2026



Tuesday’s Headlines Are a Speed Trap



Blake Aued September 22, 2026



BUSES

What Would It Take to Get ‘Fast, Free Buses’ Nationwide?



Kea Wilson September 21, 2026

BICYCLING

The Five Questions Every Community Should Ask Itself Before It Debuts a New Bike Path



Seth Lieber September 21, 2026

[See all posts →](#)

Covering the movement to end car dependency and improve biking, walking and transit in America.

Sign up for our free newsletter

Email

SIGN UP



[Staff & Board](#)

[Donate to Streetsblog](#)

[Write For Streetsblog USA or](#)

[Submit a Tip](#)

[Job Listings](#)

[Contact Us](#)

[List an event on Streetsblog](#)

[USA](#)

[NYC](#) [EMPIRE STATE](#) [MASS](#) [LA](#) [SF](#) [CAL](#) [CHI](#) [DONATE](#)

© **Copyright 2026**

